In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=pd.read_csv("C:/Users/Siddharth Singh/Downloads/telco project/Telco-Customer-Churn.csv")
```

In [17]:
```
df.head(2)
```

Out[17]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurit
0	7590-VHVEG	Female	0	Yes	No	1	No	No	DSL	N
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Ye

2 rows × 21 columns

In [18]:
```
df.isnull().sum()
```

Out[18]:
```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

In [6]:
```
df.dtypes
```

Out[6]:
```
customerID           object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges         object
Churn                object
dtype: object
```

In [14]:
```
df.head(1)
```

Out[14]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurit
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	N

1 rows × 21 columns

In [7]:
```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors='coerce')
```

In [8]:
```
df["TotalCharges"].dtype
```

Out[8]:
```
dtype('float64')
```

In [9]:
```
df["TotalCharges"].isnull().sum()
```

Out[9]:
```
np.int64(11)
```

In [10]:
```
df["TotalCharges"]=df["TotalCharges"].fillna(df["TotalCharges"].median())
```

In [11]:
```
df["TotalCharges"].dtype
```

Out[11]:
```
dtype('float64')
```

In [12]:
```
df["TotalCharges"].isnull().sum()
```

Out[12]:
```
np.int64(0)
```

In [13]:
```
df.skew(numeric_only=True)
```

Out[13]:
```
SeniorCitizen     1.833633
tenure            0.239540
MonthlyCharges   -0.220524
TotalCharges      0.963789
dtype: float64
```

In [14]:
```
df.duplicated().sum()
```

Out[14]:
```
np.int64(0)
```

In [15]:
```
df.replace("No internet service", "No", inplace=True)
df.replace("No phone service", "No", inplace=True)
```

In [16]:
```
df.dtypes
```

Out[16]:
```
customerID           object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges        float64
Churn                object
dtype: object
```

In [15]:
```
pip install sqlalchemy pyodbc
```

```
Requirement already satisfied: sqlalchemy in c:\users\siddharth singh\appdata\local\programs\python\python313\lib\site-packages (2.0.48)
Requirement already satisfied: pyodbc in c:\users\siddharth singh\appdata\local\programs\python\python313\lib\site-packages (5.3.0)
Requirement already satisfied: greenlet>=1 in c:\users\siddharth singh\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy) (3.3.2)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\siddharth singh\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy) (4.13.2)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [22]:
```
from sqlalchemy import create_engine

server = 'DESKTOP-ELR18ST'
database = 'customer_churn'
```

```
engine = create_engine(f"mssql+pyodbc://@{server}/{database}?driver=ODBC+Driver+17+for+SQL+Server")
```

In [23]:
```
df.to_sql('customer_churn', con=engine, if_exists='replace', index=False)
```

Out[23]:
```
14
```

In [ ]: